MILITARY SPECIFICATION

FABRICATED PARTS, SILICONE RUBBER (FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers rubber parts fabricated from one type of silicone rubber, for use in Aerial Mines. (see 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel

MILITARY

MIL-T-14757 - Trichlorotrifluoroethane

STANDARDS

FEDERAL

Federal Test Method STD No. 601 - Rubber, Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-417 - Rubber Compositions, Vulcanized, General Purpose, Solid

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC: 1345

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American Society for Testing and Materials Standards

D746-64T - Brittleness Temperatures of Plastics and Elastomers by Impact
D1460-60 - Change in Length of an Elastomeric Vulcanizate resulting from Immersion in a Liquid

Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.

3. REQUIREMENTS

3.1 Materials.-The rubber parts covered by this specification shall be manufactured from a silicon rubber together with added compounding ingredients of such nature and quality as to produce a vulcanized rubber conforming to the requirements which follow (see 6.4). These requirements are defined by the Code $TA5l2B_3GZZ_1Z_2$, in accordance with MIL-STD-417. The special requirements indicated by the code letter Z deals with the effect of trichlorotrifluoroethane on the rubber; Z_1 deals with the effect of diesel fuel oil on the rubber; Z_2 deals with the materials extracted from the rubber by trichlorotrifluoroethane.

3.1.1 Tensile strength.-The rubber shall have a minimum tensile strength of 1200 pounds per square inch (psi) when tested as specified in 4.3.1.

3.1.2 Durometer hardness.-The rubber shall have a Shore A Durometer hardness of 50 $\pm$ 5 when tested as specified in 4.3.2.

3.1.3 Ultimate elongation.-The rubber shall have a minimum ultimate elongation of 450 percent when tested as specified in 4.3.3.

3.1.4 Effect of heat aging.-When tested as specified in 4.3.4 the rubber shall meet the following requirements:

a. The hardness shall not increase more than 15 units from the value obtained in 4.3.2.

b. The tensile strength shall not fall below that obtained in 4.3.1 by more than 50 percent.

c. The elongation shall not fall below that obtained in 4.3.3 by more than 50 percent.

3.1.5 Low temperature brittleness.-The rubber shall pass the test specified when tested in accordance with 4.3.5.

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3.1.6 Tear strength (Code letter G).-The rubber shall have a tear strength of at least 125 pounds per square inch (psi) when tested as specified in 4.3.6.

3.1.7 Compression set (Code letter B3).-The compression set of the rubber shall not exceed 60 percent when tested as specified in 4.3.7.

3.1.8 Swell in trichlorotrifluoroethane (Code letter Z).-The rubber shall expand linearly 55 $\pm$ 5 percent within two hours when tested as specified in 4.3.8. The swollen rubber shall revert to its original dimension $\pm$ 5 percent within (1) hour when tested as specified in 4.3.8.1.

3.1.9 Swell in diesel fuel (Code letter Z_1).-The rubber shall expand linearly at least thirty (30) percent within eighteen (18) hours(s) when tested as specified in 4.3.9.

3.1.10 Material extracted by trichlorotrifluoroethane (Code letter Z_2).-The material extracted by trichlorotrifluoroethane when the rubber tested as specified in 4.3.10 shall not exceed 4.0 percent.

3.2 Workmanship.-The rubber parts shall be uniform in construction and appearance and shall be free of porous areas, bubbles and foreign matter (see 4.2.2).

3.3 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all of the quality assurance provisions, specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by code number for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required by subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 First article inspection

4.1.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.1.2.2 (see 6.2). The first article sample shall consist of the following:

a. Fifty completed parts.
b. Ten (10) 6" X 6" X .075" $\pm$ .010" sheets.
c. Two (2) 6" X 6" X .500 $\pm$ .005" sheets.

All parts and competed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in the manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

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4.1.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.2 and 4.3 of this specification and any or all requirements of the applicable drawings.

4.1.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of one or more batches of rubber parts, produced by one manufacturer in accordance with the same specification, or same specification revisions under one continuous set of operating conditions. Each batch shall consist of that quantity of rubber parts which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.2.2 Sampling for lot inspection.-From each lot a random sample of units shall be selected in accordance with Level II of MIL-STD-105, at an AQL of 1.50 percent. These units shall be examined for conformance to applicable drawings, for packing, marking, weight and workmanship as specified in the contract or purchase order.

4.2.3 Sampling for quality conformance testing.-The supplier shall perform all of the tests required to assure conformance to 3.1. For purposes of acceptance or rejection all failures to meet the applicable requirements shall be considered Major defects and in such case the lot shall be rejected. When required for Government verification, the supplier shall furnish the necessary samples molded from the same material and under the same conditions as are used in the finished product. These samples shall consist of ten (10) 6" X 6" X .075" $\pm$ .010" sheets plus two (2) 6" X 6" X .500 $\pm$ .005" sheets per lot. The samples furnished shall be suitably marked and dated for identification with the material submitted for acceptance.

4.3 Test procedures

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4.3.1 Tensile strength, major defect, defect code No. 01001.-Five (5) specimens shall be cut from 6" X 6" X .075" sheets. A die conforming to Type III of Method 4111 of Federal Test Method Standard No. 601 shall be used. The sample specimens (three or five, as required by 4111) shall be tested for conformance to 3.1.1.1 in accordance with Method 4111, of Standard No. 601.

4.3.2 Durometer A hardness, major defect, defect code No. 02001.-One (1) 6" X 6" X 0.500 ± .005" sheet shall be used to measure hardness. The hardness shall be determined using Method 3021 of Federal Test Method Standard No. 601. Five (5) readings shall be taken shifting the specimen to a new position after each reading. The median value of the results shall meet the requirement of 3.1.2.

4.3.3 Ultimate elongation, major defect, defect code 03001.-Five (5) specimens shall be prepared as in 4.3.1. These shall be tested for conformance to 3.1.3 in accordance with Method 4121 of Standard No. 601. Unless otherwise required by the contract or purchase order the elongation shall be determined on the same specimen during the same test as 4.3.1.

4.3.4 Effect of heat aging, Major defect.-Samples similar to those required for 4.3.1 and 4.3.2 shall be heated in air at a temperature of 437° ± 2°F. for 70 hours. The heat aged sample shall meet the requirements of 3.1.4 when tested as specified in 4.3.1 (defect code 04001), 4.3.2 (defect code 04002) and 4.3.3 (defect code 04003).

4.3.5 Low temperature brittleness, major defect, defect code 05001.-Five (5) specimens shall be conditioned and tested using solenoid-operated equipment in accordance with ASTM test designation D746-64T at a temperature of -67° ± 2°F. Acceptance shall be in accordance with Procedure B, Section 8 of referenced ASTM test designation.

4.3.6 Tear strength, major defect, defect code 06001.-Five (5) specimens shall be cut from a 6" X 6" X .075" sheet using a die conforming to Die B of Method 4211 of Federal Test Method Standard No. 601. The sample specimens (three or five, as required by 4211) shall be tested for conformance to 3.1.6, in accordance with Method 4211 of Standard No. 601.

4.3.7 Compression set, major defect, defect code 07001.-Two (2) specimens shall be prepared and tested in accordance with Method 3311 of Federal Test Method Standard No. 601 except that aging shall be for 70 hours at a temperature of 302° ± 2°F. The average value obatined shall meet the requirements of 3.1.7.

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4.3.8 Swelling in trichlorotrifluoroethane, major defect, defect code 08001.-Two (2) specimens of .075" thick rubber shall be cut in accordance with procedure for Apparatus B of ASTM test method designation D1460-60. The test shall be performed as specified in ASTM D1460-60, using Apparatus B with trichlorotrifluoroethane conforming to Military Specification MIL-T-14757 (MU), at a temperature of 27° ± 5°C. If the rubber does not meet the requirements of 3.1.8 for swelling in two (2) hours, continue the test for an additional hour and record results.

4.3.8.1 Recovery after swelling, major defect, defect code 08002.-After the test for swelling has been completed, remove the samples from the fluid and allow to stand on the specimen holder in a well-ventilated room. At the end of one (1) hour, measure the change in length of the specimens. If the specimens fail to meet any requirement of 3.1.8 the lot shall be rejected.

4.3.9 Swell in diesel fuel, major defect, defect code 09001.-Two (2) specimens shall be prepared as specified for use in Apparatus A of ASTM test method D1460-60. These shall be immersed in No. 1 diesel oil conforming to VV-F-800. The average value of swelling shall determine conformance to the requirement of 3.1.9.

4.3.10 Material extracted by trichlorotrifluoroethane, major defect, defect code No. 10001.-A composite sample shall be prepared from test units in accordance with Method 1211 of Federal Test Method Standard No. 601. A 2 gram sample shall be tested for conformance to 3.1.10 using Method 16311, of Federal Test Method Standard No. 601, except that the extraction mixture shall be replaced by trichlorotrifluoroethane.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level C.-The rubber parts shall be packed in conformance with manufacturer's commercial practice so as to provide protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2 Marking.-Marking shall be as specified in the contract or purchase order and shall include the following as a minimum:

Manufacturer
Manufacturer's identification of the material
Date of manufacture
Lot number and quantity
Contract or order number

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6. NOTES

6.1 Intended Use.-The material covered by this specification is intended for use in Aerial Mines.

6.2 Ordering data.-Procurement documents shall specify title, number and date of this document, and provisions for submission of First Article Samples.

6.3 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 A satisfactory rubber has been prepared using silicone rubber resin SE555U, made by the General Electric Company, Silicone Products Department, Waterford, New York 12188, and vulcanizing agent Cadox TS-50 made by the Cadet Chemical Co., Burt, New York. Rubber prepared from any other materials shall be approved by Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND8, prior to use.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1345-A-253

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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